SALES & PROFITABILITY ANALYSIS

Power BI Dashboard Project Report

Prepared by	Atharva Shinde
Tool	Microsoft Power BI Desktop, Power Query, DAX
Dataset	PowerBI_Sales_Profitability_Dataset.csv
Reporting period	January–May 2026
Dataset size	40 rows · 11 columns · practice dataset

Project purpose: transform retail sales records into an interactive report to review sales, profitability, regional performance, product results, and discount patterns.

Portfolio practice project. Add final dashboard screenshots and verify all findings before publishing.

1. Executive Summary

This project uses Microsoft Power BI to explore a small retail sales dataset. The dashboard is designed to summarize sales and profit, compare performance across regions and product categories, identify negative-profit records, and explore how discount rates relate to profitability. Interactive filters allow users to narrow analysis by order date, region, and category.

Key findings — complete after validating your dashboard:

• [Finding 1: highest-sales region and its sales value]

• [Finding 2: most profitable category/subcategory and its profit]

• [Finding 3: loss-making product or observed discount/profit pattern]

Overall recommendation: [State the most important action supported by your results.]

2. Business Problem and Objectives

Management needs an accessible way to understand where sales are generated, which products contribute to profit, and where losses may require investigation. An interactive dashboard makes it easier to compare regions, categories, and time periods.

• Measure total sales, total profit, distinct orders, quantity sold, and profit margin.

• Compare sales across regions and profit across categories and subcategories.

• Inspect product-level performance and identify negative-profit observations.

• Explore discount levels alongside profit without assuming correlation proves causation.

• Communicate findings and practical recommendations through visuals and a concise report.

3. Dataset Description

The practice dataset contains 40 rows and 11 columns. It is a small synthetic learning dataset and should not be interpreted as real company data or as statistically representative of retail performance.

Column	Meaning	Suggested type
Order ID	Order identifier	Text
Order Date	Date of the order	Date
Region / State	Geographic grouping	Text
Category / Sub-Category	Product classification	Text
Product Name	Product description	Text
Quantity	Units sold	Whole number
Sales	Sales amount	Decimal/currency
Discount	Discount rate; 0.05 = 5%	Decimal/percentage
Profit	Profit or loss amount	Decimal/currency

Data quality checks: verify data types, check blank values and errors, confirm discounts are stored as decimals, and validate that totals respond correctly to slicers. Do not remove duplicate-looking rows unless they are confirmed duplicates.

4. Tools and Methodology

• Tools: Power BI Desktop, Power Query, DAX, and CSV.

• Import the CSV dataset into Power BI.

• Use Power Query to check types, blanks, errors, and data consistency.

- Create DAX measures for core business KPIs.
- Build Executive Overview and Product & Profit Analysis pages.
- Test slicers and validate visual totals against the source data.
- Interpret the dashboard and document evidence-based findings.

5. Key DAX Measures

Measure	DAX expression	Purpose
Total Sales	SUM('SalesData'[Sales])	Sum of sales amounts.
Total Profit	SUM('SalesData'[Profit])	Sum of profit/loss amounts.
Total Orders	DISTINCTCOUNT('SalesData'[Order ID])	Number of distinct order IDs.
Total Quantity	SUM('SalesData'[Quantity])	Total units sold.
Profit Margin	DIVIDE([Total Profit], [Total Sales], 0)	Profit divided by sales.
Average Discount	AVERAGE('SalesData'[Discount])	Simple average of discount rates.
Loss-Making Rows	CALCULATE(COUNTROWS('SalesData'), 'SalesData'[Profit] < 0)	Rows with negative profit.

Replace 'SalesData' with your actual Power BI table name. Format Profit Margin and Average Discount as percentages. Avoid summing discount rates in table totals.

6. Dashboard Design

Page 1 — Executive Overview

- KPI cards: Total Sales, Total Profit, Profit Margin, and Total Orders.
- Bar chart: Total Sales by Region.
- Column chart: Total Profit by Category.
- Line chart: Total Sales over time, preferably grouped by month.
- Slicers: Order Date, Region, and Category.

Page 2 — Product & Profit Analysis

- Table: Product Name, Category, Sales, Profit, and Discount.
- Bar chart: Total Profit by Product Name.
- Bar chart: Total Profit by Sub-Category.
- Discount analysis: scatter chart with individual products/observations, or average discount by subcategory.
- Slicers: Region and Category.

Insert a legible screenshot of each completed dashboard page before publishing. Make sure screenshots show the final, corrected report.

7. Key Findings and Analysis

Fill in the following table using actual values and patterns visible in Power BI. These prompts are intentionally blank so unverified results are not presented as facts.

Area	Evidence to record	Interpretation prompt
Regional performance	Highest-sales region: [region]; sales: [value].	What might explain the regional difference?
Category profitability	Most profitable category: [category]; profit: [value].	What strategy merits further review?

Area	Evidence to record	Interpretation prompt
Loss-making products	Product/subcategory: [name]; profit: [value].	Review pricing, costs, and discounts.
Monthly sales trend	Observed trend: [months and values].	Consider implications for planning.
Discount analysis	Observed relationship: [describe chart].	Investigate further; correlation does not prove causation.

8. Business Recommendations

• Regional strategy: review high-performing regions and assess whether successful practices can transfer elsewhere.

• Product strategy: compare product-level profit and sales before deciding which products to promote or review.

• Loss management: investigate negative-profit records, including pricing, costs, returns, and discounts where data is available.

• Discount policy: evaluate discount levels alongside sales and profit; gather more evidence before changing policy.

• Monitoring: use the dashboard to review performance periodically and investigate unusual changes.

Keep only recommendations supported by actual dashboard findings and available data.

9. Limitations

• The dataset contains only 40 practice records, so results are illustrative rather than statistically robust.

• The dataset does not include cost breakdowns, customer segments, returns, or marketing spend, limiting causal interpretation.

• The observed discount/profit relationship does not establish that discounts cause lower profit.

• Conclusions depend on slicer selections and measure definitions.

10. Conclusion

This project demonstrates a basic analytics workflow using Power BI: preparing data, creating DAX measures, designing interactive visualizations, and translating results into business questions and recommendations. Add actual dashboard screenshots and evidence-based findings before using the report as portfolio documentation.

Final deliverables: Power BI .pbix file, CSV dataset, final dashboard screenshots, and this report. Complete the findings and insert screenshots before publishing.